

Hasan Haider

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Work Experience

GeoAnalysis Engineering - FrontEnd React Developer

Jul 2023 – Jun 2024

React, Tailwind CSS, JavaScript, Flowbite, Node.js

- Developed a fully responsive website from client requirements using modern frontend technologies.
- Gathered requirements, refactored the legacy codebase, and built a parallel site that replaced the original.
- Improved performance and loading time by **55%** while optimizing layouts for multiple screen sizes.

GAE Internship - Software Engineering Intern

May 2025 – Aug 2025

Python, PyInstaller, Git, Debugging

- Refactored the Intelligent Thermal Conductivity Meter from the ground up, improving modularity and reducing loading time by **38%**.
- Reduced active threads by **25%** through cleaner threading logic and optimized runtime behavior.
- Built a lightweight, thread-safe live data plotter for real-time instrument readings.
- Created automated error handling and a PyInstaller pipeline for generating easy-to-use Windows installers.

Projects

Ollama-Based Multimedia Edubot

Jan 2026 – Apr 2026

Python, Wikimedia API, ChromaDB, Ollama, SQL, LLMs, Taiga

- Led a team of 6 people to add multimedia capabilities to an educational chatbot by retrieving relevant Wikimedia images instead of generating AI images, reducing operational costs.
- Built a database in ChromaDB to cache those images for faster retrieval.
- Added the ability to run the whole system offline using Ollama and added a switch for Online/Offline mode.
- Coordinated planning using Taiga board, weekly scrum meetings and other agile methodologies.

Unity based FPS shooter with procedural world generation

Sep 2025 – Dec 2025

C#, Unity

- Built a 3D 6DoF FPS shooter game in Unity with procedural world generation, where the world and level change dynamically every time it's played.
- Added an enemy AI system with variable difficulty that would track and shoot at the player in 3d space while avoiding all moving objects.

Discord.js Web Bot

Dec 2019 – Jan 2021

JavaScript, discord.js, Selenium, Node.js

- Built an automated Discord bot that scrapes external websites with Selenium and displays requested data in servers through user commands.

IoT-Based Weighing Scale

Jan 2020 – Mar 2020

Python, Arduino, ESP8266

- Modified a standard weighing scale to wirelessly send weight data to Adafruit IO for remote dashboard viewing.

Education

M.S. Software Engineering, Cybersecurity Specialization

Aug 2024 – May 2026

Arizona State University, Tempe, AZ

B.Tech Information Technology

Aug 2019 – Jul 2023

KIET Group of Institutions, Delhi NCR, India GPA: 7.32/10

Skills

Languages: Python, C/C++, C#, JavaScript, SQL, HTML/CSS

AI Tools: OpenAI Codex, Google Antigravity, Gemini, Cursor

Frontend: React, Tailwind CSS, Flowbite

Backend & Tools: Node.js, PyInstaller, Selenium, Git, Arduino, ESP8266, Unity, Linux

Other: OCR with Tesseract, DeepL API, Neural Networks, Real-Time Plotting, Thread Safety, LLMs, CUDA

Journal Publications

- Rizvi Z.H., Akhtar S.J., Husain S.M.B., Khan M., **Haider, H.**, Naqvi S., Tirth V., Wuttke F. (2022). Neural network approaches for computation of soil thermal conductivity. *Mathematics*, 10(21), 3957.
- Rizvi, Z.H., Baqir Husain, S.M., **Haider, H.**, Wuttke, F. (2020). Effective thermal conductivity of sands estimated by Group Method of Data Handling. *Materials Today: Proceedings*, 26(2), 2103–2107.
- Rizvi, Z.H., Akhtar, S.J., **Haider, H.**, Follmann, J., Wuttke, F. (2019). Estimation of seismic wave velocities of metamorphic rocks using artificial neural network. *Materials Today: Proceedings*, 26(2), 324–330.